STAT 400 Homework 4: Random Variables, Expected Value, and Variance

Instructions. Show your work and simplify final answers. You may use:

$$\mathbb{E}[aX+b] = a\mathbb{E}[X]+b, \qquad \text{Var}(aX+b) = a^2\text{Var}(X), \qquad \text{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$$

If random variables are stated to be independent, you may also use:

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

1) **Three step experiment (mixture distribution).**
A bag contains 10 coins: 9 fair coins (type F) and 1 unfair coin (type U). For the unfair coin, $P(H) = 0.8$.
   a) Step 1: Choose one coin uniformly at random from the bag.
   b) Step 2: Toss the chosen coin once.
   c) Step 3: If the toss is H, roll a fair 4-sided die with faces $\{1,2,3,4\}$. If the toss is T, roll a 4-sided die with

$$P(1) = 0.2, \qquad P(2) = 0.3 \qquad P(3) = 0.2 \qquad P(4) = 0.3$$

   Define events $F = \{\text{chosen coin is fair}\}$ and $G = \{\text{coin toss is heads}\}$. Let D be the die result from Step 3, and define the indicator random variable

$$X = \mathbf{1}\{D \text{ is even}\}$$

   a) Compute $P(F)$, $P(G)$, and $P(G|F)$.

$$P(F) = \frac{9}{10} = \mathbf{0.9}$$

$$P(G \mid F) = \frac{1}{2} = \mathbf{0.5}$$

$$P(G) = P(F)P(G \mid F) + P\left(F^{\complement}\right)P\left(G|F^{\complement}\right)$$
$$P(G) = \frac{9}{10}\cdot\frac{1}{2} + \frac{1}{10}\cdot\frac{8}{10}$$
$$P(G) = \frac{9}{20} + \frac{8}{100}$$
$$P(G) = \frac{53}{100} = \mathbf{0.53}$$

   b) Compute $P(D=1)$, $P(D=2)$, $P(D=3)$, and $P(D=4)$.

$$P(D=1) = P(G)P(1|G) + P\left(G^{\complement}\right)P\left(1|G^{\complement}\right)$$
$$P(D=1) = \frac{53}{100}\cdot\frac{1}{4} + \left(1-\frac{53}{100}\right)\cdot\frac{2}{10}$$
$$P(D=1) = \mathbf{0.2265}$$

1) b)

$$P(D=2) = P(G)P(2|G) + P\left(G^{\complement}\right)P\left(2|G^{\complement}\right)$$
$$P(D=2) = \frac{53}{100}\cdot\frac{1}{4} + \left(1-\frac{53}{100}\right)\cdot\frac{3}{10}$$
$$P(D=2) = \mathbf{0.2735}$$

$$P(D=3) = P(G)P(3|G) + P\left(G^{\complement}\right)P\left(3|G^{\complement}\right)$$
$$P(D=3) = \frac{53}{100}\cdot\frac{1}{4} + \left(1-\frac{53}{100}\right)\cdot\frac{2}{10}$$
$$P(D=3) = \mathbf{0.2265}$$

$$P(D=4) = P(G)P(4|G) + P\left(G^{\complement}\right)P\left(4|G^{\complement}\right)$$
$$P(D=4) = \frac{53}{100}\cdot\frac{1}{4} + \left(1-\frac{53}{100}\right)\cdot\frac{3}{10}$$
$$P(D=4) = \mathbf{0.2735}$$

c) Compute $\mathbb{E}[D]$ and $\text{Var}(D)$.

$$\mathbb{E}[D] = 1\cdot 0.2265 + 2\cdot 0.2735 + 3\cdot 0.2265 + 4\cdot 0.2735$$
$$\mathbb{E}[D] = \mathbf{2.547}$$

$$\mathbb{E}\left[D^2\right] = 1^2\cdot 0.2265 + 2^2\cdot 0.2735 + 3^2\cdot 0.2265 + 4^2\cdot 0.2735$$
$$\mathbb{E}\left[D^2\right] = 7.735$$

$$\text{Var}(D) = \mathbb{E}\left[D^2\right] - (\mathbb{E}[D])^2$$
$$\text{Var}(D) = 7.735 - 2.547^2$$
$$\text{Var}(D) = \mathbf{1.247791}$$

d) Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

$$\mathbb{E}[X] = 0\cdot 0.2265 + 1\cdot 0.2735 + 0\cdot 0.2265 + 1\cdot 0.2735$$
$$\mathbb{E}[X] = \mathbf{0.547}$$

$$\mathbb{E}\left[X^2\right] = 0^2\cdot 0.2265 + 1^2\cdot 0.2735 + 0^2\cdot 0.2265 + 1^2\cdot 0.2735$$
$$\mathbb{E}\left[X^2\right] = 0.547$$

$$\text{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$$
$$\text{Var}(X) = 0.547 - 0.547^2$$
$$\text{Var}(X) = \mathbf{0.247791}$$

1) e) Determine whether X and F are independent (justify using probabilities).

$$
\begin{aligned}
P(X \cap F) &= P(X)P(F) \\
P(X \mid F)P(F) &= P(X)P(F) \\
P(X \mid F) &= P(X) \\
P(X \mid F) &= P(D = 2) + P(D = 4) \\
P(X \mid F) &= 0.2375 + 0.2375 \\
P(X \mid F) &= 0.475
\end{aligned}
$$

$$
P(G|F)P(2|G) + P\left(G^{\complement}|F\right)P\left(2|G^{\complement}\right) +
$$

$$
P(G|F)P(4|G) + P\left(G^{\complement}|F\right)P\left(4|G^{\complement}\right) = 0.475
$$

$$
0.5 \cdot 0.25 + 0.5 \cdot 0.3 + 0.5 \cdot 0.25 + 0.5 \cdot 0.3 = 0.475
$$

$$
0.55 \neq 0.475
$$

X and F are dependent.

2) **Four coin tosses (binomial random variables).**
A coin with $P(H) = 0.5$ is tossed 4 times. Let H be the number of heads and T be the number of tails.

a) State the distribution of H and compute $\mathbb{E}[H]$ and $\text{Var}(H)$.

$$\boldsymbol{H \sim \text{Binom}(4, 0.5)}$$

$$\mathbb{E}[H] = 4 \cdot 0.5 = \mathbf{2}$$

$$\text{Var}(H) = 4 \cdot 0.5 \cdot (1 - 0.5) = \mathbf{1}$$

b) Compute $\mathbb{E}[T]$ and $\text{Var}(T)$ using the relationship between H and T.

$$\mathbb{E}[T] = \mathbb{E}[4 - H] = 4 - \mathbb{E}[H] = \mathbf{2}$$

$$\text{Var}(T) = \text{Var}(4 - H) = \text{Var}(H) = \mathbf{1}$$

c) Define $Y = \mathbf{1}\{H > T\}$. Compute $\mathbb{E}[Y]$ and $\text{Var}(Y)$.

$$\mathbb{E}[Y] = 1 \cdot P(H = 3) + 1 \cdot P(H = 4)$$
$$\mathbb{E}[Y] = \binom{4}{3}(0.5)^4 + \binom{4}{4}(0.5)^4$$
$$\mathbb{E}[Y] = 0.25 + 0.0625$$
$$\mathbb{E}[Y] = \mathbf{0.3125}$$

$$\mathbb{E}\left[Y^2\right] = \binom{4}{3}(0.5)^4 + \binom{4}{4}(0.5)^4$$
$$\mathbb{E}\left[Y^2\right] = 0.25 + 0.0625$$
$$\mathbb{E}\left[Y^2\right] = 0.3125$$
$$\text{Var}(Y) = \mathbb{E}\left[Y^2\right] - (\mathbb{E}[Y])^2$$
$$\text{Var}(Y) = 0.3125 - 0.3125^2$$
$$\text{Var}(Y) = \mathbf{0.21484375}$$

3) **Four biased coin tosses (compare to the fair case).**
A coin with $P(H) = 0.9$ is tossed 4 times. Let H be the number of heads and T be the number of tails.

a) State the distribution of H and compute $\mathbb{E}[H]$ and $\text{Var}(H)$.

$$\boldsymbol{H \sim \text{Binom}(4, 0.9)}$$

$$\mathbb{E}[H] = 4 \cdot 0.9 = \mathbf{3.6}$$

$$\text{Var}(H) = 4 \cdot 0.9 \cdot (1 - 0.9) = \mathbf{0.36}$$

b) Compute $\mathbb{E}[T]$ and $\text{Var}(T)$.

$$\mathbb{E}(T) = 4 \cdot 0.1 = \mathbf{0.4}$$
$$\text{Var}(T) = 4 \cdot 0.1 \cdot (1 - 0.1) = \mathbf{0.36}$$

c) Define $Y = \mathbf{1}\{H > T\}$. Compute $\mathbb{E}[Y]$ and $\text{Var}(Y)$.

$$\mathbb{E}[Y] = 1 \cdot P(H = 3) + 1 \cdot P(H = 4)$$
$$\mathbb{E}[Y] = \binom{4}{3}(0.9)^3(0.1)^1 + \binom{4}{4}(0.9)^4$$
$$\mathbb{E}[Y] = 0.2916 + 0.6561$$
$$\mathbb{E}[Y] = \mathbf{0.9477}$$

$$\mathbb{E}\left[Y^2\right] = 1^2 \cdot \binom{4}{3}(0.9)^3(0.1)^1 + 1^2 \cdot \binom{4}{4}(0.9)^4$$
$$\mathbb{E}\left[Y^2\right] = 0.2916 + 0.6561$$
$$\mathbb{E}\left[Y^2\right] = 0.9477$$
$$\text{Var}(Y) = \mathbb{E}\left[Y^2\right] - (\mathbb{E}[Y])^2$$
$$\text{Var}(Y) = 0.9477 - 0.9477^2$$
$$\text{Var}(Y) = \mathbf{0.04956471}$$

d) Briefly compare your answers in this problem to Problem 2. Which quantities changed, and why?

Every answer changed due to the bias toward the head side.
- **$\mathbb{E}[H]$ decreased from 2 to 3.6, and $\mathbb{E}[T]$ decreased from 2 to 0.4.**
- **$\text{Var}(H)$ and $\text{Var}(T)$ decreased from 1 to 0.36.**
- **$\mathbb{E}[Y]$ increased from 0.3125 to 0.9477, and $\text{Var}(Y)$ decreased from 0.21484375 to 0.04956471.**

4) **Two fair dice (sum, max, difference).**
Two fair six-sided dice are rolled. Let the outcomes be (X_1, X_2).

Define the random variables

$$S = X_1 + X_2, \qquad M = \max\{X_1, X_2\}, \qquad W = X_1 - X_2$$

a) Compute $\mathbb{E}[S]$ and $\mathrm{Var}(S)$.

$$\begin{aligned}\mathbb{E}[S] &= 2 \cdot P(S=2) + 3 \cdot P(S=3) + 4 \cdot P(S=4) + 5 \cdot P(S=5) \\ &\quad +6 \cdot P(S=6) + 7 \cdot P(S=7) + 8 \cdot P(S=8) + 9 \cdot P(S=9) \\ &\quad +10 \cdot P(S=10) + 11 \cdot P(S=11) + 12 \cdot P(S=12)\end{aligned}$$

$$\begin{aligned}\mathbb{E}[S] &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + 5 \cdot \frac{4}{36} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{6}{36} \\ &\quad +8 \cdot \frac{5}{36} + 9 \cdot \frac{4}{36} + 10 \cdot \frac{3}{36} + 11 \cdot \frac{2}{36} + 12 \cdot \frac{1}{36}\end{aligned}$$

$$\mathbb{E}[S] = \mathbf{7}$$

$$\begin{aligned}\mathbb{E}\left[S^2\right] &= 2^2 \cdot \frac{1}{36} + 3^2 \cdot \frac{2}{36} + 4^2 \cdot \frac{3}{36} + 5^2 \cdot \frac{4}{36} + 6^2 \cdot \frac{5}{36} + 7^2 \cdot \frac{6}{36} \\ &\quad +8^2 \cdot \frac{5}{36} + 9^2 \cdot \frac{4}{36} + 10^2 \cdot \frac{3}{36} + 11^2 \cdot \frac{2}{36} + 12^2 \cdot \frac{1}{36}\end{aligned}$$

$$\mathbb{E}\left[S^2\right] = 54.833$$

$$\mathrm{Var}(S) = \mathbb{E}\left[S^2\right] - (\mathbb{E}[S])^2$$

$$\mathrm{Var}(S) = 54.833 - (7)^2$$

$$\mathrm{Var}(S) = \mathbf{5.833}$$

b) Compute $\mathbb{E}[W]$ and $\mathrm{Var}(W)$.

$$\begin{aligned}\mathbb{E}[W] &= -5 \cdot P(W=-5) - 4 \cdot P(W=-4) - 3 \cdot P(W=-3) - 2 \cdot P(W=-2) - 1 \cdot P(W=-1) \\ &\quad +0 \cdot P(W=0) + 1 \cdot P(W=1) + 2 \cdot P(W=2) + 3 \cdot P(W=3) + 4 \cdot P(W=4) + 5 \cdot P(W=5)\end{aligned}$$

$$\mathbb{E}[W] = -5 \cdot \frac{1}{36} - 4 \cdot \frac{2}{36} - 3 \cdot \frac{3}{36} - 2 \cdot \frac{4}{36} - \frac{5}{36} + 0 + \frac{5}{36} + 2 \cdot \frac{4}{36} + 3 \cdot \frac{3}{36} + 4 \cdot \frac{2}{36} + 5 \cdot \frac{1}{36}$$

$$\mathbb{E}[W] = \mathbf{0}$$

$$\begin{aligned}\mathbb{E}\left[W^2\right] &= (-5)^2 \cdot \frac{1}{36} + (-4)^2 \cdot \frac{2}{36} + (-3)^2 \cdot \frac{3}{36} + (-2)^2 \cdot \frac{4}{36} + (-1)^2 \frac{5}{36} \\ &\quad +0 + 1^2 \cdot \frac{5}{36} + 2^2 \cdot \frac{4}{36} + 3^2 \cdot \frac{3}{36} + 4^2 \cdot \frac{2}{36} + 5^2 \cdot \frac{1}{36}\end{aligned}$$

$$\mathbb{E}\left[W^2\right] = 5.833$$

$$\mathrm{Var}(W) = \mathbb{E}\left[W^2\right] - (E[W])^2$$

$$\mathrm{Var}(W) = 5.833 - 0^2$$

$$\mathrm{Var}(W) = \mathbf{5.833}$$

4) c) Compute $\mathbb{E}[M]$.

$$\mathbb{E}[M] = 1 \cdot P(M = 1) + 2 \cdot P(M = 2) + 3 \cdot P(M = 3) + 4 \cdot P(M = 4) + 5 \cdot P(M = 5) + 6 \cdot P(M = 6)$$

$$\mathbb{E}[M] = \frac{1}{36} + 2 \cdot \frac{3}{36} + 3 \cdot \frac{5}{36} + 4 \cdot \frac{7}{36} + 5 \cdot \frac{9}{36} + 6 \cdot \frac{11}{36}$$

$$\mathbb{E}[M] = \mathbf{4.472}$$

d) Compute $\text{Var}(M)$.

$$\mathbb{E}\left[M^2\right] = 1^2 \cdot \frac{1}{36} + 2^2 \cdot \frac{3}{36} + 3^2 \cdot \frac{5}{36} + 4^2 \cdot \frac{7}{36} + 5^2 \cdot \frac{9}{36} + 6^2 \cdot \frac{11}{36}$$

$$\mathbb{E}\left[M^2\right] = 21.972$$

$$\text{Var}(M) = \mathbb{E}\left[M^2\right] - (\mathbb{E}[M])^2$$

$$\text{Var}(M) = 24.472 - 4.472^2$$

$$\text{Var}(M) = \mathbf{1.971}$$